LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Quick Revision: Heredity

Key terms

Term	Meaning
Heredity	Passing of traits from parents to offspring
Variation	Differences among individuals of a species; very little in asexual reproduction, more in sexual reproduction
Gene	A section of DNA that carries the information for a protein, and so for a trait
Dominant trait	Expressed even if only one copy is present (T)
Recessive trait	Expressed only when both copies are recessive (tt)
Genotype / phenotype	Gene make-up (TT, Tt, tt) / visible trait (tall, short)
F_1, F_2	First and second filial (offspring) generations

Mendel's experiments (garden pea)

- Why pea: many clear contrasting traits, short life cycle, self-pollinating but easy to cross, many seeds.
- **Monohybrid cross:** TT (tall) × tt (short) → F_1 all tall (Tt). $F_1 \times F_1 \rightarrow F_2$: TT, Tt, Tt, tt.
- F_2 ratios: **phenotype 3 tall : 1 short**; **genotype 1 TT : 2 Tt : 1 tt**.
- The short trait is not lost in F_1. It is hidden by the dominant trait and reappears in F_2.
- **Dihybrid cross:** RRYY (round yellow) × rryy (wrinkled green) → F_1 all RrYy (round yellow).
- F_2: **9 round yellow : 3 round green : 3 wrinkled yellow : 1 wrinkled green**, so traits are inherited **independently**.

Checkerboard for Tt × Tt. Gametes from each parent: T and t.
T × T = TT (tall) · T × t = Tt (tall) · t × T = Tt (tall) · t × t = tt (short)

How genes control traits

Gene → protein (often an enzyme) → trait. Example: an efficient enzyme makes more growth hormone, so the plant is tall. An altered gene gives a less efficient enzyme, less hormone, and a short plant. Each parent gives **one copy** of each gene to the offspring through the germ cells.

Sex determination in humans

- Humans have 23 pairs of chromosomes. One pair is the sex chromosomes.
- Mother: XX, so every egg carries X. Father: XY, so half the sperm carry X and half carry Y.
- X sperm + egg → XX (girl). Y sperm + egg → XY (boy). So the **father's sperm decides** the sex. The chance is 50 : 50.
- In some animals the environment decides sex: in some reptiles, the temperature at which eggs are kept. Some snails can change sex.

Common mistakes

- Writing the F_2 ratio as 1 : 2 : 1 when the question asks for the **phenotypic** ratio (3 : 1).
- Saying the mother decides the sex of the child.
- Forgetting to show gametes in a cross. They carry marks!

Quick check

1. In a cross Tt × tt, what fraction of offspring will be short?

2. Why is the sex of a child decided by the father?